

Customer Sales Analysis Report

CUSTOMER SALES ANALYSIS REPORT

Total Revenue: \$12,365,048

Total Customers: 500

Average Order Value: \$123,650

Top Customer: CUST016 - \$373,932

1. Project Overview

This project analyzes customer purchasing patterns and evaluates overall sales performance. The primary objective is to identify highly valuable customers, understand regional and seasonal revenue distributions, and evaluate customer retention rates to provide actionable business recommendations.

2. Setup Instructions

- Create a project directory and place the datasets (sales_data.csv, customer_churn.csv) inside.
- Initialize a virtual environment: `python-mvenv venv`
- Activate the environment: `. env\Scripts ctivate`
- Install dependencies via requirements.txt: `pip install pandas matplotlib seaborn jupyter`

3. Code Structure

- customer_analysis.ipynb: Main execution notebook containing data manipulation.
- sales_data.csv & customer_churn.csv: Raw data sources.
- requirements.txt: Dependency list.
- analysis_report.pdf : Final documentation.

4. Technical Details & Algorithms

Utilized **pandas** for all transformations. Date strings were converted to datetime objects. Customer ID strings were stripped of prefixes ('CUST', 'C') and cast to integers to facilitate a clean inner join between the sales and churn datasets. Executed aggregations and generated a pivot table mapping product revenue across regions.

5. Testing Evidence & Validation

Verified both datasets contained 0 missing values. Confirmed data integrity by testing if the provided `Total_Sales` strictly matched the calculated `Quantity * Price` for all rows.

6. Visual Documentation

(Note: See attached dashboard generated via `matplotlib` and `seaborn` depicting Regional Revenue, Seasonal Trends, Product Revenue, and Retention Rate).

7. Core Analysis & Business Insights

1. Who are our most valuable customers?

Our top-performing customers by revenue generation are ID C016 (\$373,932), C007 (\$363,870), and C083 (\$350,888).

Recommendation: Enroll these top-tier clients into a VIP loyalty program.

2. What products sell best together?

A market basket analysis of the current data reveals a 1-to-1 customer-to-purchase ratio in this dataset snapshot.

Recommendation: Track recurring customer transactions over a longer timeline to map cross-selling.

3. Which regions have the highest sales?

The North region is leading (\$3,983,635), followed by the South region (\$3,737,852).

Recommendation: Replicate North region marketing strategies in the underperforming West region (\$2,123,922).

4. What are the seasonal trends?

Sales experience high volatility in Q1, with a notable dip in Month 2 followed by a massive peak in Month 3 (\$4,485,006).

Recommendation: Ensure inventory for top revenue drivers is heavily stocked prior to Month 3.

5. How can we improve customer retention?

Our current retention rate is 89.4%, meaning our churn rate is roughly 10.6%.

Recommendation: Implement automated exit surveys for churning customers and offer targeted discounts to at-risk accounts.